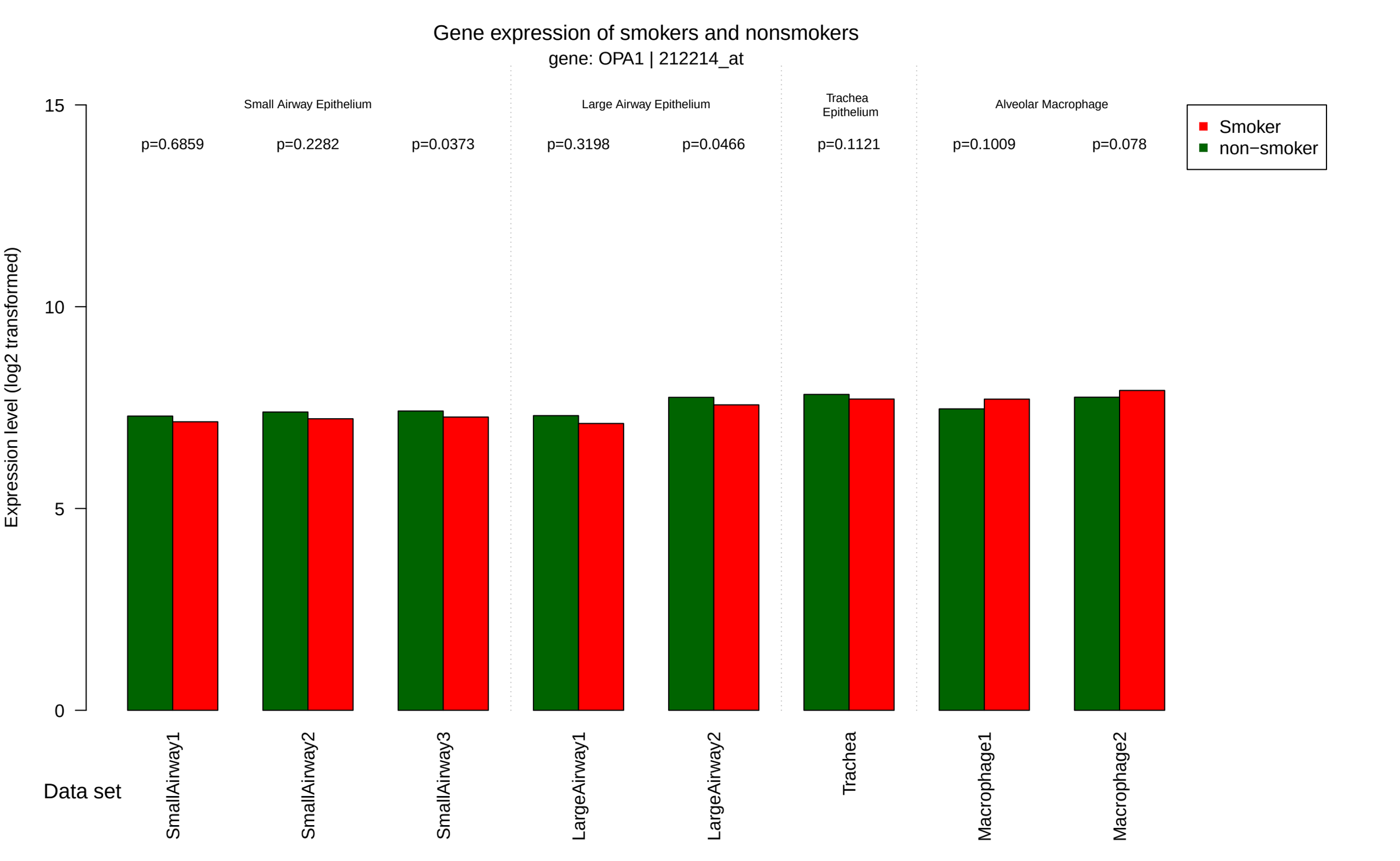
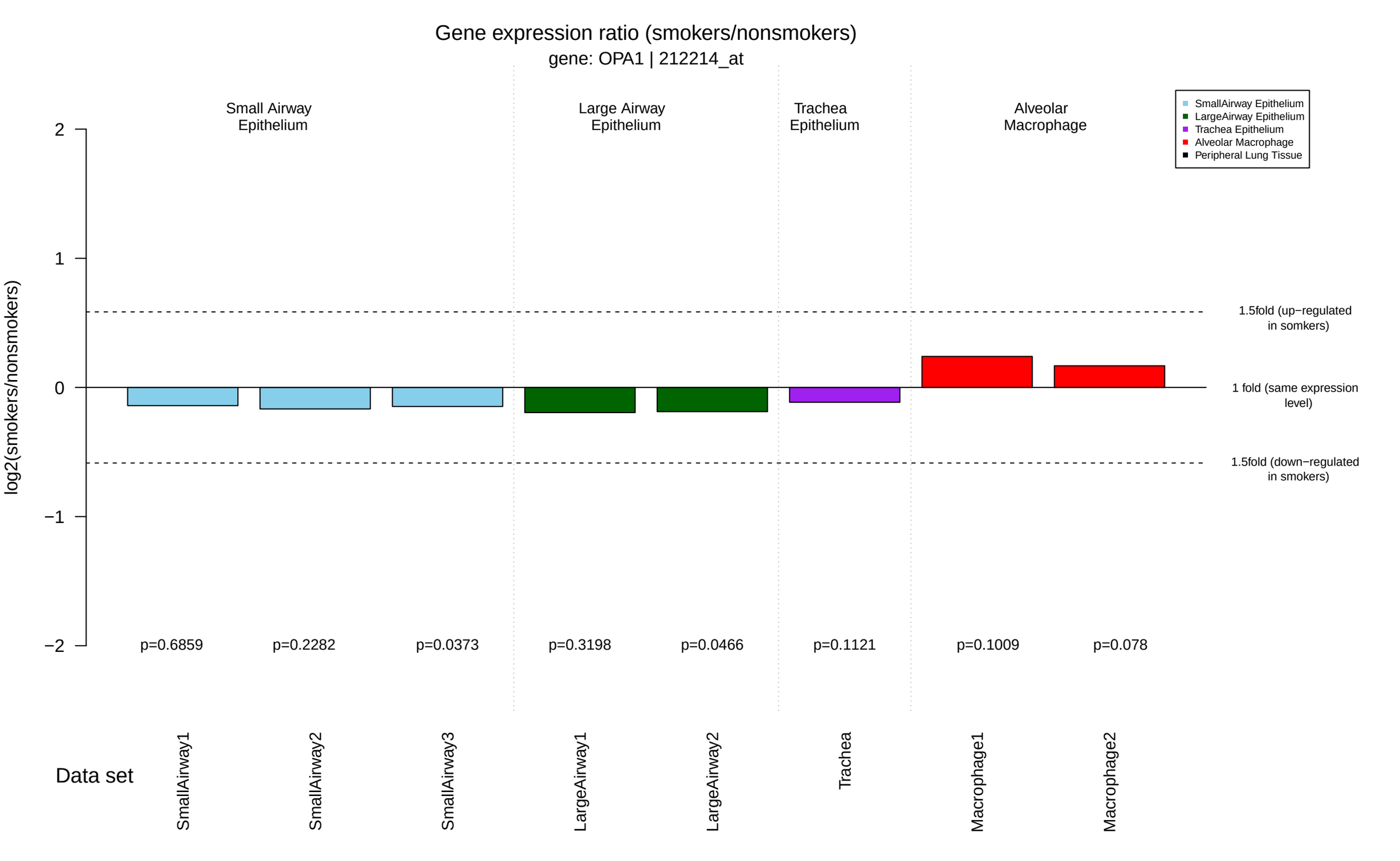
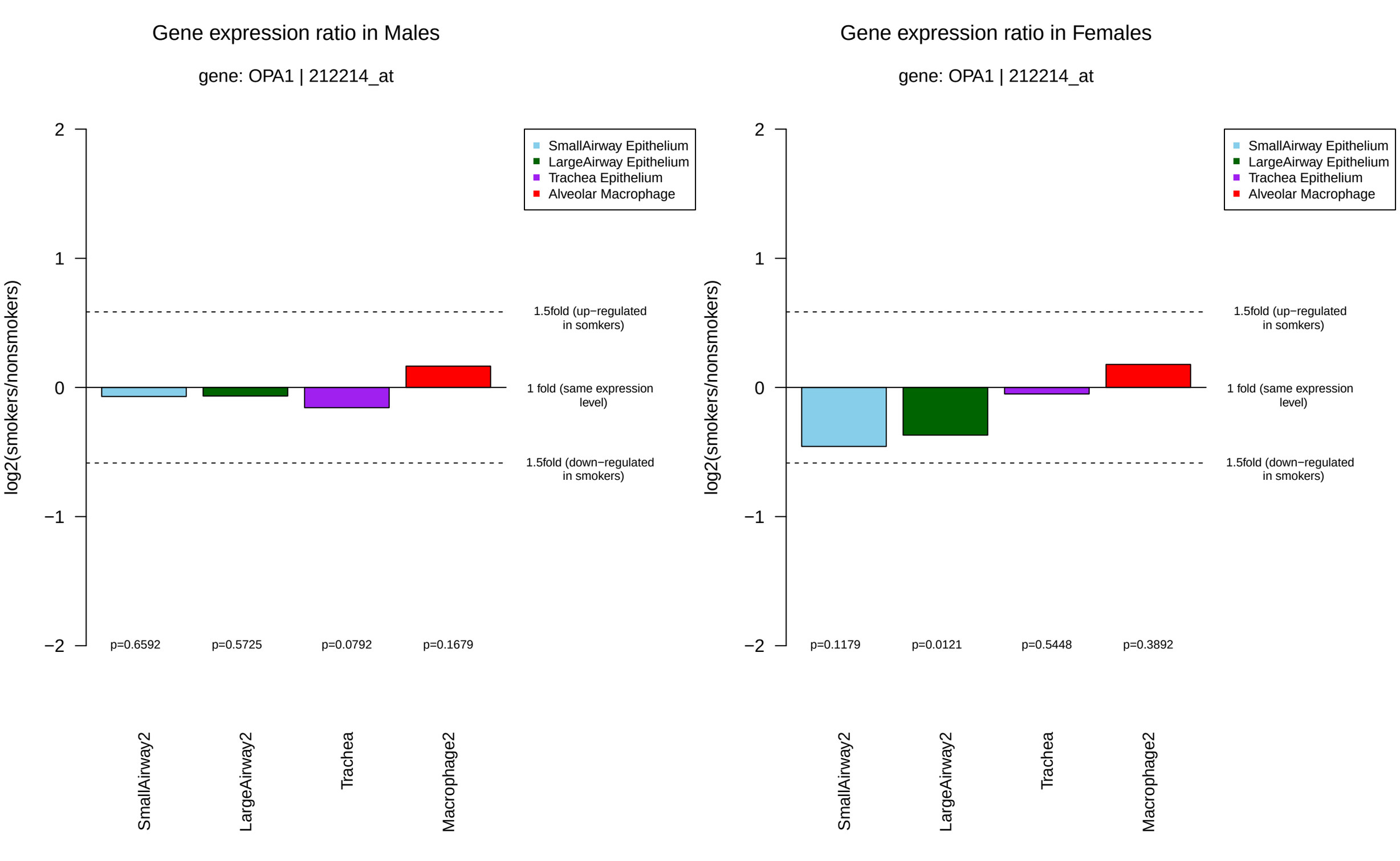
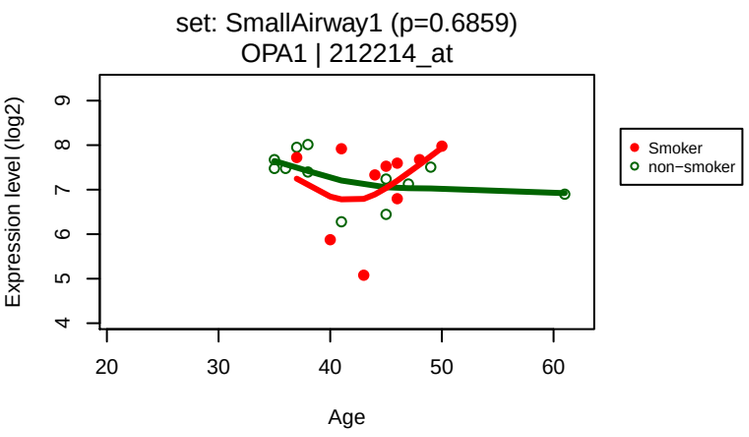
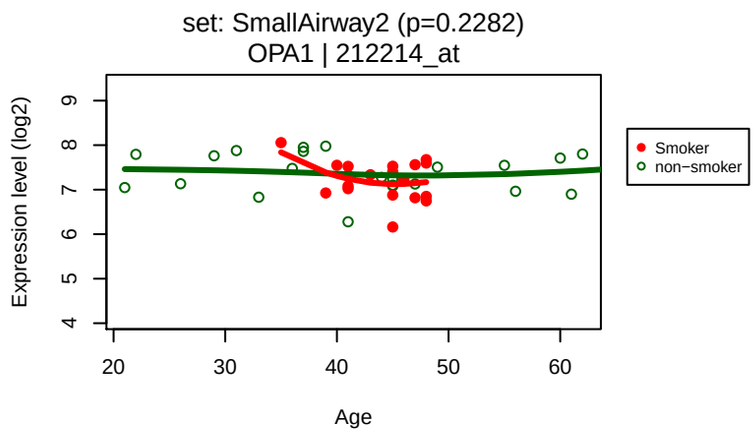
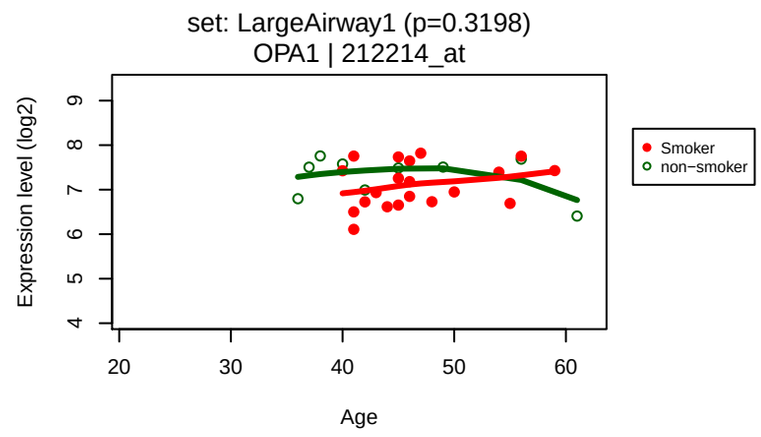

Gene expression of smokers and nonsmokers
gene: OPA1 | 212214_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6859
p=0.2282
p=0.0373
p=0.3198
p=0.0466
p=0.1121
p=0.1009
p=0.078
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

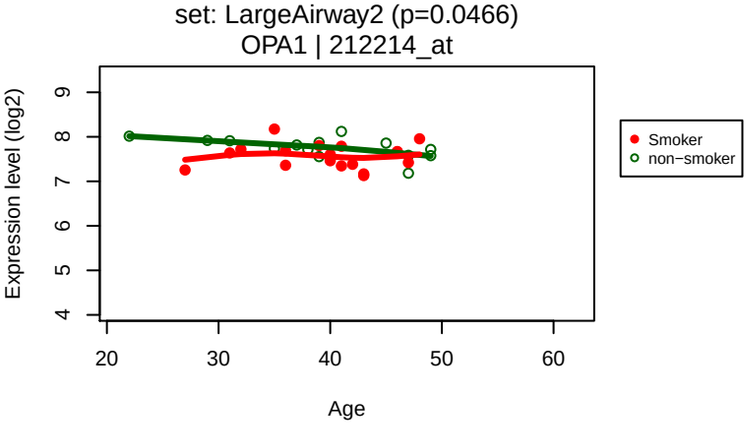

Gene expression ratio (smokers/nonsmokers)
gene: OPA1 | 212214_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6859
p=0.2282
p=0.0373
p=0.3198
p=0.0466
p=0.1121
p=0.1009
p=0.078
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

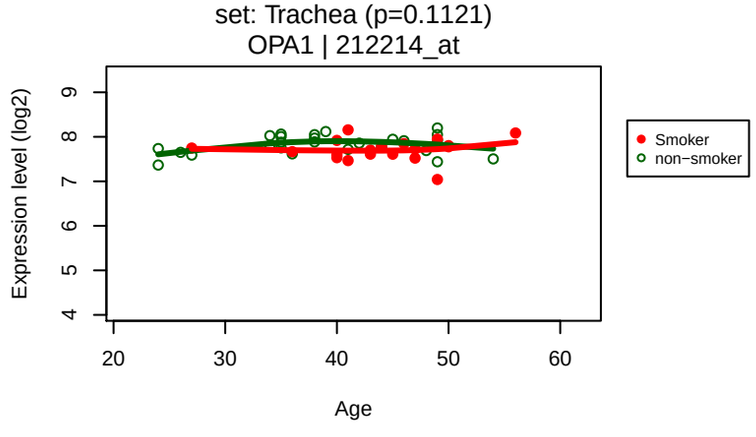

Gene expression ratio in Males
gene: OPA1 | 212214_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6592
p=0.5725
p=0.0792
p=0.1679
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OPA1 | 212214_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1179
p=0.0121
p=0.5448
p=0.3892
SmallAirway2
LargeAirway2
Trachea
Macrophage2

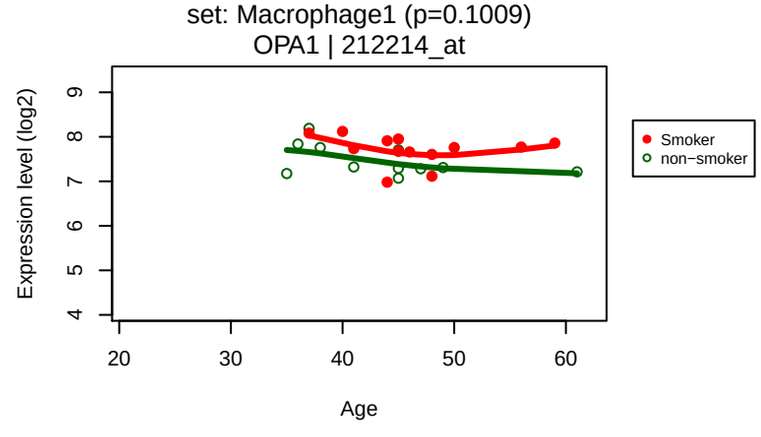

set: SmallAirway1 (p=0.6859)
OPA1 | 212214_at
Expression level (log2)
Age
Smoker
non−smoker

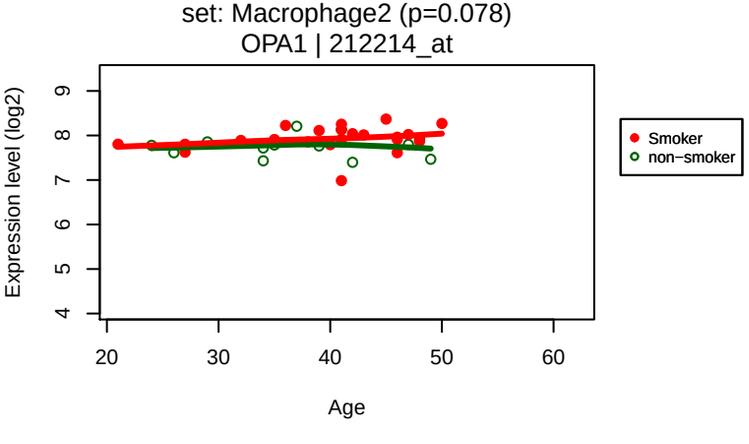

set: SmallAirway2 (p=0.2282)
OPA1 | 212214_at
Expression level (log2)
Age
Smoker
non−smoker

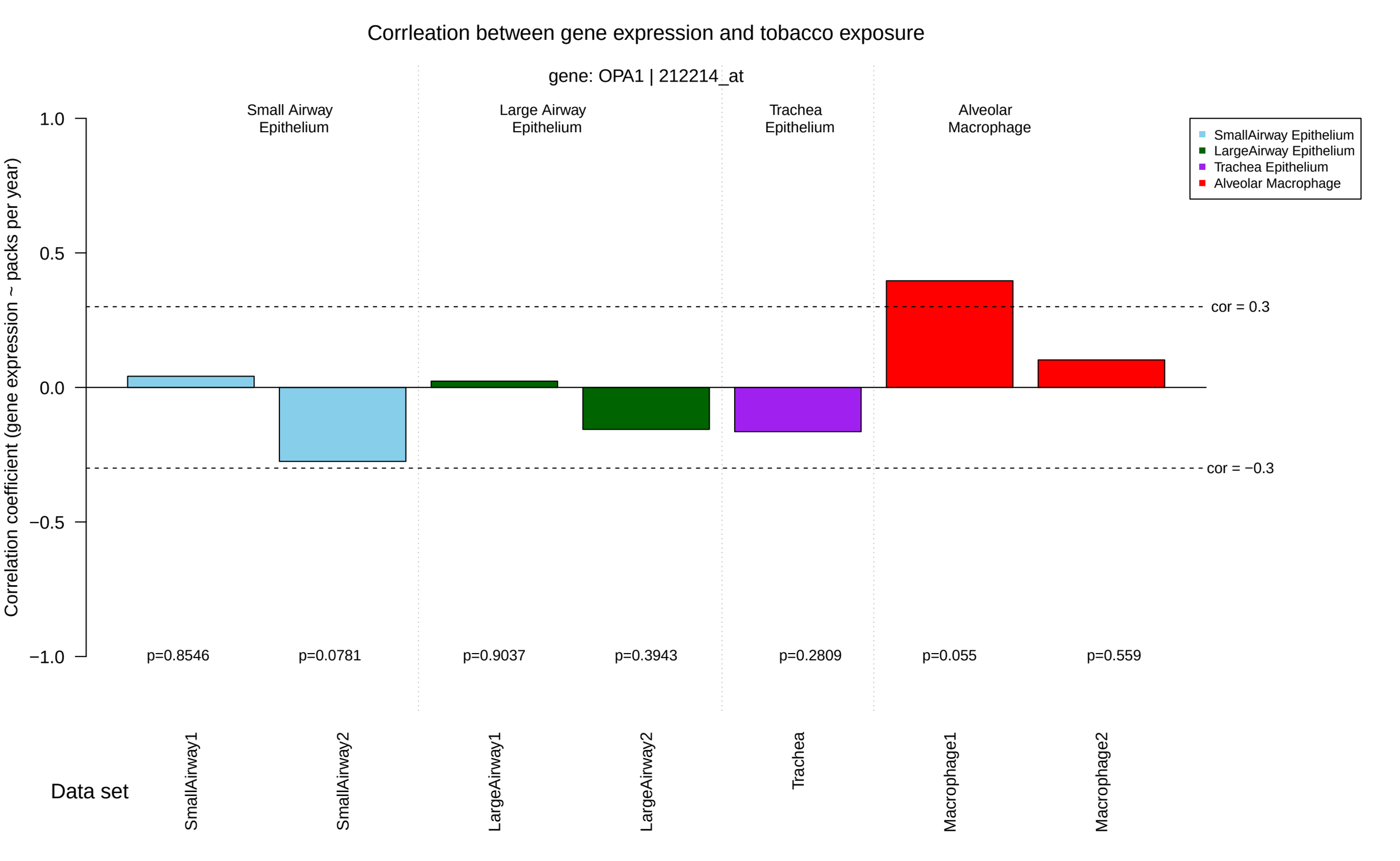

set: LargeAirway1 (p=0.3198)
OPA1 | 212214_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0466)
OPA1 | 212214_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1121)
OPA1 | 212214_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1009)
OPA1 | 212214_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.078)
OPA1 | 212214_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OPA1 | 212214_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8546
p=0.0781
p=0.9037
p=0.3943
p=0.2809
p=0.055
p=0.559
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2